A countable-block overcomplete set for an ordinal interval space

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Status. Candidate partial result, likely valid. This packet proves the first countable-block case left open by the earlier lane-16 packet: in ZFC, $C([0, \omega_1 \cdot \omega])$ admits an overcomplete set. It does not claim the full $\xi < \omega_2$ solution to Koszmider's Question 48.

1 Source question

The source paper is:

> P. Koszmider, *On the existence of overcomplete sets in some classical nonseparable Banach spaces*, arXiv:2006.00806.

On page 25, Question 48 asks:

> Can one prove in ZFC that the Banach spaces $C([0, \xi])$ for all ordinals $\xi < \omega_2$ admit overcomplete sets?

classes for Banach spaces like the following three questions:

Question 48. *Can one prove in ZFC that the Banach spaces $C([0, \xi])$ for all ordinals $\xi < \omega_2$ admit overcomplete sets?*

Partial results related to Question 48 are Theorem 11 (2) and result of [37] (Theorem 5 (3) in this paper). They cover all remaining ordinals as $C([0, \xi])$ admits a fundamental biorthogonal system $(1_{[0,\eta]}, \delta_\eta - \delta_{\eta+1})_{\eta<\xi}$ and $1_{[0,\xi]}, \delta_\xi$.

2 Definitions and source tools

Definition 1. *Let X be a Banach space. A set $D \subseteq X$ is* overcomplete *if $|D| = \mathrm{dens}(X)$ and every subset $A \subseteq D$ with $|A| = |D|$ has dense linear span in X.*

We use two standard facts used explicitly in Koszmider's paper. First, because $[0, \omega_1]$ is scattered, every functional on $C([0, \omega_1])$ is an atomic measure with countable support. Second, there are almost coherent injections $e_\alpha : \alpha \to \mathbb{N}$ ($\alpha < \omega_1$) such that

$$e_\beta \restriction \alpha =^* e_\alpha \qquad (\alpha < \beta < \omega_1),$$

and they may be chosen with $\mathbb{N} \setminus e_\alpha[\alpha]$ infinite. As in the source proof, this implies the following exact stabilization: if $A \subseteq \omega_1$ is uncountable and $\gamma < \omega_1$, then there are an uncountable $A' \subseteq A \setminus (\gamma + 1)$ and an injection $g : \gamma + 1 \to \mathbb{N}$ such that

$$e_\alpha \restriction (\gamma + 1) = g \qquad (\alpha \in A').$$

3 Proof Intuition

The previous partial packet stopped before $\omega_1 \cdot \omega$ because the finite-codimensional countable-support criterion does not survive countably many ω_1-blocks. The obstacle is cancellation: a functional on a countable c_0-sum is a summable sequence of block functionals, and the block evaluations can cancel each other.

The construction below prevents that cancellation by assigning each block its own sparse set of Taylor exponents. On an uncountable subfamily, the coherent injections make the value of each fixed block functional an analytic power series in one real parameter r_α. Since different blocks use disjoint exponent sets, an analytic identity forces the coefficients of every block separately to vanish. A small endpoint-detector series with unbounded coefficients then kills the endpoint mass in each copy of $C([0, \omega_1])$, just as in the strengthened $C([0, \omega_1])$ construction from the earlier run work.

4 The countable-block theorem

Theorem 1. *Let*

$$Y = C([0, \omega_1]).$$

Then the Banach space

$$X = \mathbb{R} \oplus \Big(\bigoplus_{n<\omega} Y \Big)_{c_0}$$

admits a set $\{x_\alpha : \alpha < \omega_1\}$ such that every nonzero functional on X vanishes on at most countably many x_α's. In particular, X admits an overcomplete set.

Proof. Write $K = [0, \omega_1]$, $u_\alpha = \mathbf{1}_{[0,\alpha]} \in C(K)$ ($\alpha < \omega_1$), and $\mathbf{1} = \mathbf{1}_K$. In this proof $\mathbb{N} = \{0, 1, 2, \ldots\}$.

Choose pairwise disjoint infinite sets

$$P_\infty, P_0, P_1, \ldots \subseteq \mathbb{N}$$

and enumerations $p_j : \mathbb{N} \to P_j$, $j \in \{\infty, 0, 1, \ldots\}$, such that

$$p_j(k) \geq 2k + 1 \qquad (k < \omega).$$

Put

$$q_j(t) = \sum_{k=0}^{\infty} 2^k t^{p_j(k)} \qquad (|t| < 1/\sqrt{2}).$$

Fix $0 < b < 1/2$, and choose pairwise distinct $r_\alpha \in (b/2, b)$ for $\alpha < \omega_1$. Also fix positive scalars $a_n \downarrow 0$, for instance $a_n = 2^{-n-1}$.

For $j < \omega$ and $\alpha < \omega_1$, define $v_\alpha^{(j)} \in Y$ by

$$v_\alpha^{(j)} = r_\alpha^{p_j(0)} u_\alpha + \sum_{\beta<\alpha} r_\alpha^{p_j(e_\alpha(\beta)+1)} (u_\beta - u_\alpha) + q_j(r_\alpha)\mathbf{1}.$$

The series converges absolutely in Y, because e_α is injective and $\sum_k b^{p_j(k)} \le \sum_k b^{2k+1} < \infty$. The vectors $v_\alpha^{(j)}$ are uniformly bounded in j and α. Hence

$$x_\alpha = \big(q_\infty(r_\alpha), (a_n v_\alpha^{(n)})_{n<\omega}\big) \in X$$

is well-defined. Since q_∞ has positive coefficients, it is strictly increasing on $(0, b)$, so the x_α's are pairwise distinct.

We prove the functional separation property. Let

$$\Phi = (c, (\mu_n)_{n<\omega}) \in \mathbb{R} \oplus \Big(\bigoplus_{n<\omega} Y^*\Big)_{\ell_1} = X^*$$

and suppose that $\Phi(x_\alpha) = 0$ for all α in some uncountable set $A \subseteq \omega_1$. For each n, the measure μ_n has countable support in K; since there are only countably many n's, the union of the supports below ω_1 is bounded. Choose $\gamma < \omega_1$ such that

$$\bigcup_{n<\omega} \big(\operatorname{supp}(\mu_n) \setminus \{\omega_1\}\big) \subseteq [0, \gamma].$$

By exact stabilization, pass to an uncountable $A' \subseteq A \setminus (\gamma + 1)$ and an injection $g : \gamma + 1 \to \mathbb{N}$ such that $e_\alpha \restriction (\gamma + 1) = g$ for every $\alpha \in A'$.

For $n < \omega$, put

$$b_{n,\beta} = \mu_n(u_\beta) \quad (\beta \le \gamma), \qquad d_n = \mu_n(\mathbf{1}),$$

and let s_n be the eventual value of $\mu_n(u_\xi)$ for $\xi > \gamma$. For $\alpha \in A'$, with $t = r_\alpha$, the definition of $v_\alpha^{(n)}$ gives

$$\mu_n(v_\alpha^{(n)}) = s_n t^{p_n(0)} + \sum_{\beta\le\gamma} (b_{n,\beta} - s_n) t^{p_n(g(\beta)+1)} + d_n q_n(t).$$

Consequently the analytic function

$$F(t) = c\, q_\infty(t) + \sum_{n=0}^{\infty} a_n \left[s_n t^{p_n(0)} + \sum_{\beta\le\gamma} (b_{n,\beta} - s_n) t^{p_n(g(\beta)+1)} + d_n q_n(t) \right]$$

vanishes at every r_α with $\alpha \in A'$. The series defining F converges normally on compact subdiscs of $|t| < 1/\sqrt{2}$, because $(\|\mu_n\|)_n \in \ell_1$, the a_n's are bounded, and the displayed series for each block is bounded by a constant depending only on b times $\|\mu_n\|$. The set $\{r_\alpha : \alpha \in A'\}$ has an accumulation point in $(b/2, b)$, so the identity theorem gives $F \equiv 0$.

Compare Taylor coefficients. The sets $P_\infty, P_0, P_1, \ldots$ are pairwise disjoint. The coefficients on P_∞ are $c\, 2^k$; hence $c = 0$. Fix $n < \omega$. Since $a_n > 0$, the coefficients on P_n vanish. The range of g has infinite complement, so there are infinitely many k's outside

$$\{0\} \cup \{g(\beta) + 1 : \beta \le \gamma\}.$$

For such k, the coefficient of $t^{p_n(k)}$ is $a_n d_n 2^k$, so $d_n = 0$. The coefficient of $t^{p_n(0)}$ then gives $s_n = 0$. Finally, for each $\beta \le \gamma$, the coefficient of $t^{p_n(g(\beta)+1)}$ gives $b_{n,\beta} = 0$.

Thus $\mu_n(\mathbf{1}) = 0$, $\mu_n(u_\beta) = 0$ for all $\beta \le \gamma$, and $\mu_n(u_\xi) = s_n = 0$ for all $\xi > \gamma$. The linear span of $\mathbf{1}$ and the u_ξ's is dense in $C([0,\omega_1])$: it is a unital algebra separating points of the compact ordinal interval. Therefore $\mu_n = 0$. Since this holds for every n, and $c = 0$, we have $\Phi = 0$.

We have proved that any functional vanishing on an uncountable subfamily of $\{x_\alpha : \alpha < \omega_1\}$ must be zero. Since $\operatorname{dens} X = \omega_1$, it follows that every uncountable subfamily has dense linear span by Hahn–Banach. Hence the family is overcomplete in X. □

Theorem 2. *In ZFC, $C([0,\omega_1\cdot\omega])$ admits an overcomplete set.*

Proof. Let $K = [0,\omega_1\cdot\omega]$, and put $t_\infty = \omega_1\cdot\omega$. For $n < \omega$, let

$$I_0 = [0,\omega_1], \qquad I_n = (\omega_1\cdot n, \omega_1\cdot(n+1)] \quad (n \ge 1).$$

Then $K \setminus \{t_\infty\}$ is the topological disjoint union of the compact-open pieces I_n, and each I_n is homeomorphic to $[0,\omega_1]$. The map

$$Tf = \Big(f(t_\infty), \big(f|_{I_n} - f(t_\infty)\mathbf{1}_{I_n}\big)_{n<\omega}\Big)$$

is a linear isomorphism from $C(K)$ onto

$$\mathbb{R} \oplus \Big(\bigoplus_{n<\omega} C([0,\omega_1])\Big)_{c_0}.$$

Indeed, continuity at t_∞ is exactly the condition that the displayed block sequence tends to zero in norm. Apply Theorem 1 and pull the overcomplete set back through T. □

5 Limitations and novelty check

This improves the earlier lane-16 packet for ordinal intervals below $\omega_1\cdot\omega$ by adding the first missing countable-block endpoint $\xi = \omega_1\cdot\omega$.

The proof is not yet a full answer to Question 48. It handles a countable c_0-sum of copies of $C([0,\omega_1])$ by assigning countably many blocks disjoint Taylor supports. The same simple coefficient separation does not directly handle ordinal intervals whose natural block structure involves ω_1-many interacting ω_1-blocks, such as the ω_1^2-level.

Local indexes were searched for arXiv:2006.00806, ordinal-interval overcomplete keywords, and the previous packet slug. A fresh web/literature pass checked the exact Question 48 wording, ordinal $C([0,\alpha])$ classification phrases, Bessaga–Pelczynski/Kislyakov/Gul'ko–Os'kin classification references, and $C([0,\omega_1\cdot\omega])$ isomorphism phrases. This found classification literature and related operator theory on ordinal $C(K)$ spaces, but no later paper explicitly resolving Question 48 or this countable-block subcase.

6 Human review recommendation

Review as a likely valid partial result. The main points to check are:

1. the coefficient-separation argument using pairwise disjoint Taylor exponent sets P_n;
2. the normal convergence of the analytic function F;

3. the use of exact stabilization for one countable bound γ that simultaneously covers all block measures;
4. the identification of $C([0, \omega_1 \cdot \omega])$ with the scalar plus countable c_0-sum model.

References

[1] P. Koszmider, *On the existence of overcomplete sets in some classical nonseparable Banach spaces*, J. Funct. Anal. **281** (2021), no. 9, 109172; arXiv:2006.00806.

[2] T. Russo and J. Somaglia, *Overcomplete sets in non-separable Banach spaces*, Proc. Amer. Math. Soc. **149** (2021), 701–714; arXiv:1912.08690.